



Use of pre-workout dietary supplements is associated with lower sleep duration among adolescents and young adults

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ABSTRACT

This study aimed to investigate the association between the use of pre-workout dietary supplements and average sleep duration among adolescents and young adults. Data from Wave 2 of the Canadian Study of Adolescent Behaviors ($N = 912$) were analyzed. Multinomial logistic regression analyses were conducted to investigate the cross-sectional association between the use of pre-workout dietary supplements in the past 12 months and average sleep duration in the past 2 weeks. Adolescents and young adults who reported pre-workout dietary supplement use were at greater risk (RRR 2.53, 95% CI 1.27–5.05, $p = 0.09$) of reporting ≤ 5 h of average sleep per night relative to 8 h of sleep. Findings underscore that the high caffeine content of pre-workout dietary supplements is associated with shorter sleep durations, which are well below the recommended guidelines for adolescents and young adults. Health and mental health care professionals should educate pre-workout users on the association between use and sleep.

1. Introduction

Pre-workout drinks and powders (hereafter referred to as "pre-workout") are multi-ingredient dietary supplements [1] that are widely used among young people. According to some estimates, 25 % of adolescents and young adults have reported using pre-workouts in the past 12 months [2]. Pre-workout supplements are typically composed of several ingredients, including caffeine, branched-chain amino acids, nitrates, creatine, and sweeteners [1,3]. The combination of these ingredients is purported to enhance exercise performance [4], with some evidence suggesting improvements in muscular endurance, but little effect on endurance exercise performance and strength [1].

Caffeine is a popular sports and performance aid due to its ergogenic effects [5]. Consequently, caffeine content per serving in pre-workout is high. Recent research has documented that the caffeine content ranges from 91 to 387 mg per serving of pre-workout [6], with an average of 254 mg per serving [3]. However, caffeine consumption can negatively impact sleep, including a significant reduction in sleep duration [7].

It is recommended that adolescents and young adults sleep between 7 and 10 h per night [8,9], which is critical for their physiological, psychological, and social health and development [10]. Despite this recommendation, many young people sleep significantly less than the

recommended amount [11]. To date, no known studies have investigated the association between pre-workout use and sleep duration among adolescents and young adults, which was the aim of this study. It was hypothesized that pre-workout use would be associated with lower sleep duration.

2. Methods

Data were from participants who completed Wave 2 ($N = 912$) of the Canadian Study of Adolescent Health Behaviors, a study of adolescents and young adults ages 16–30 years across Canada. The initial sample ($N = 2774$) was recruited in November and December 2021 via Instagram and Snapchat advertisements, with no specific targeting criteria applied except for age and country. Wave 2 data were collected one year later (November and December 2022). Participants from Wave 1 who elected to participate in Wave 2 were contacted via email. The surveys from Wave 1 and Wave 2 were nearly identical, with only minor differences, including the addition of the sleep duration item. The data were collected online via Qualtrics, and participants were eligible to enter a draw for one of three Apple iPads in recognition of their participation. To prevent survey infiltration by bots, several recommended techniques were added to the survey, including reCAPTCHA

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verification, attention checks, and honeypot items. Additionally, several Qualtrics features were utilized, including “prevent multiple submissions” (i.e., preventing participants from taking the survey multiple times) and “block indexing options” (i.e., blocking search engines from including the survey in their search results). Ethics approval was obtained from the Health Sciences Research Ethics Board at the University of Toronto (#41707), and written informed consent was obtained from all participants.

2.1. Measures

Pre-workout use (any/none) in the past 12 months was assessed based on the question, “Over the past 12 months, have you used any of the following appearance- and performance-enhancing dietary supplements and substances?” with “pre-workout drinks or powders” as a potential option [2].

Average hours of sleep per night over the past two weeks were assessed based on self-report using the question, “In the past two weeks, how many hours per night did you sleep on average?” Response options increased incrementally by one hour, ranging from “0 h” to “15 or more hours.” Responses were categorized into 5 or fewer hours, 6 h, 7 hours, 8 h (base category), and 9 or more hours. Eight hours of sleep per night was chosen as the base category, as this number of hours aligns with the recommended sleep guidelines for both adolescents (8–10 h) and adults (7–9 h) in Canada [8,9].

Sociodemographic variables included self-reported age, race/ethnicity, sexual orientation, and highest completed education. Gender included those who identified as cisgender girls and women, cisgender boys and men, and transgender and gender-expansive participants. Control variables included symptoms of depression and anxiety. Depression was measured using the Patient Health Questionnaire 9-Item (range: 0–27) [12]. Anxiety was measured using the Generalized Anxiety Disorder 7-Item (range: 0–21) [13]. Weight training (any/none) in the past 30 days was also assessed.

2.2. Statistical analysis

Descriptive statistics were used to characterize the sample. One multinomial logistic regression analysis was conducted to determine the association between pre-workout use in the past 12 months and average nightly sleep duration, adjusting for the sociodemographic and control variables. Analyses were not stratified by age, given the overall small number of participants 18 years and younger ($n = 34$), which limited analytic power. Multicollinearity was checked and indicated no concerns (all VIFs < 2), except for PHQ-9 and GAD-7 scores (VIFs 3.7 and 3.8, respectively), given that depression and anxiety symptoms often co-occur. Additionally, effect modification by gender was assessed, and no statistically significant findings were observed. Therefore, analyses were not stratified by gender. Statistical analyses were conducted using Stata 17.0, and statistical significance was determined using two-sided $p < 0.05$.

3. Results

The sample was demographically diverse, with a mean age of 23.4 years ($SD = 3.8$), 57.4 % identifying as girls and young women, 62.2 % identifying as White, and 55.1 % identifying as heterosexual (Table 1). The majority of participants reported averaging 7 h of sleep per night over the past 2 weeks (34.9 %). Over one in five participants reported pre-workout use in the past 12 months (22.2 %).

The multinomial logistic regression model was statistically significant compared with the null model, $\chi^2(60) = 161.4$, $p < 0.001$. The model demonstrated acceptable fit (Pseudo $R^2 = 0.07$). Results from multinomial logistic regression analyses revealed significant associations between pre-workout use and shorter sleep duration (Table 2). Pre-workout use in the past 12 months, compared to no use, was

Table 1
Sample Characteristics of Participants in the Canadian Study of Adolescent Health Behaviors ($N = 912$).

	%/ M (SD)
Age	23.4 (3.8)
Gender	
Girls and Young Women	57.4
Boys and Young Men	33.8
Transgender/Gender Expansive	8.8
Race/ethnicity	
White	62.2
Black	3.5
Asian	17.1
Other	6.2
Multi-Racial	11.0
Sexual Orientation	
Heterosexual	55.1
Gay/Lesbian	9.2
Bisexual	17.6
Queer, Questioning, Other	18.1
Highest Completed Education	
High School Diploma or Less	33.4
Undergraduate or College Diploma	48.4
Master's Degree or Higher	18.2
Depression Symptoms ^a	2.5 (1.8)
Anxiety Symptoms ^b	2.4 (1.8)
Any Weight Training, Past 30 Days	50.6
Average Sleep per Night, Past 2 Weeks	
≤ 5 H	10.6
6 H	25.0
7 H	34.9
8 H	20.4
≥ 9 H	9.0
Pre-Workout Use, Past 12 Months	22.2

^a Measured using the Patient Health Questionnaire 9-Item (range = 0–27).
^b Measures using the Generalized Anxiety Disorder 7-Item (range = 0–21)
M = Mean; SD = Standard deviation.

significantly associated with 5 or fewer hours of sleep, relative to 8 h of sleep.

4. Discussion

This study is among the first to investigate the association between pre-workout use and sleep duration among a diverse sample of adolescents and young adults in Canada. Confirming our hypothesis, the use of pre-workout in the past 12 months was associated with significantly fewer average hours of sleep per night in the past 2 weeks.

Pre-workout supplements are marketed and used to increase acute exercise performance [4] and, therefore, contain high concentrations of caffeine [3], a known ergogenic aid [5]. However, caffeine consumption can negatively impact sleep, including reducing sleep duration [7]. The negative impacts of caffeine on sleep may be attributed to its neuro-physiological effects. Specifically, caffeine blocks adenosine receptors, which promotes wakefulness, as well as delays the endogenous melatonin rhythm [14]. There is also evidence that caffeine activates the sympathetic nervous system (e.g., increases heart rate and blood pressure), which may disrupt sleep [15,16]. Given these effects, a recent meta-analysis recommended that pre-workout supplements (with an average of 217 mg of caffeine per serving) be consumed more than 13 h before bedtime [7]. Adolescents and young adults are likely limited in the time of day they can work out, such as after educational and/or occupational responsibilities (i.e., in the afternoon or evening), which would place their consumption of pre-workout well within the recommended 13 h. Thus, the effects of caffeine on the body may impair sleep duration.

Despite the novel findings of this study, several limitations should be noted. The sample was collected via non-probability methods (i.e., social media), and participants self-selected their involvement in Wave 2

Table 2

Associations Between Pre-Workout Use and Average Sleep per Night among Participants in the Canadian Study of Adolescent Health Behaviors.

	Average Sleep per Night, Past 2 Weeks							
	≤ 5 H vs. 8 H		6 Hours vs. 8 H		7 Hours vs. 8 H		≥ 9 H vs. 8 H	
	RRR (95 % CI) ^a	p	RRR (95 % CI) ^a	p	RRR (95 % CI) ^a	p	RRR (95 % CI) ^a	p
Pre-Workout Use, Past 12 Months	2.53 (1.27–5.05)	0.009	1.33 (0.77–2.31)	0.309	1.09 (0.66–1.81)	0.727	1.26 (0.59–2.69)	0.546

Note: The table presents the abbreviated outputs of a single multinomial logistic regression analysis with pre-workout use as the independent variable and average sleep per night as the dependent variable, adjusting for covariates.

Boldface indicates statistical significance at $p < 0.05$.

RRR = Relative risk ratio; CI = Confidence interval.

^a Adjusted for age, gender, race/ethnicity, sexual orientation, highest completed education, PHQ-9 score, GAD-7 score, and any weight training in the past 30 days.

of data collection, which limits the generalizability of the findings. Specifically, there was relatively high attrition between Waves, and there is a gender and age imbalance that should be addressed in future research. However, the sample was demographically diverse and represented 12 of the 13 provinces and territories in Canada (except for Prince Edward Island). Future research should replicate this investigation using nationally representative data. The binary pre-workout use item does not provide details on how the dosage, frequency, and time of pre-workout use, as well as the various formulations of pre-workout supplements, may impact sleep duration. This is a critical area of future investigation that may inform pre-workout use instructions and guidelines for healthcare professionals to support users. Similarly, the single-item assessment of sleep duration alone does not provide a deeper understanding of how pre-workout use may impact overall sleep quality. Indeed, future investigation is needed to better characterize the association between patterns of pre-workout use and sleep quality. Although we adjusted for any weight training in the past 30 days, this binary measure may not capture critical data that could explain shorter sleep durations among adolescents and young adults. This includes whether weight training displaces sleep due to waking up early or going to bed late for exercise purposes. We also lacked information on the current employment and schooling status of participants, which may have implications for their sleep patterns. Finally, it is possible that there is a bidirectional relationship between pre-workout use and lower sleep duration. That is, individuals may not sleep enough and therefore use pre-workout to provide sufficient energy to work out. Investigating the temporal relationship between pre-workout use and sleep duration is warranted.

5. Conclusion

This study documents that the use of pre-workout supplements is associated with lower sleep duration among a sample of adolescents and young adults. Given the high prevalence of pre-workout use among these populations and the importance of sleep for their health and development, the findings have important implications. Health and mental health care professionals should provide education and guidance on the potential adverse effects that pre-workout use can have on sleep duration if consumed too close to bedtime. For instance, they should advise adolescents and young adults, particularly athletes, to avoid consuming pre-workout supplements 12 to 14 h before bedtime [7]. This aligns with recommendations to increase knowledge among supplement users to reduce possible harms [17]. Health and mental health care professionals can also provide adolescents and young adults with sleep hygiene information, particularly given the importance of sleep on athletic performance and muscle building [18]. Future investigation is warranted to gather more information on how the frequency and dose of pre-workout use impact sleep duration and quality.

Ethics approval

Ethics approval in line with the principles of the Declaration of

Helsinki was obtained from the research ethics board at the University of Toronto (#41707), and written informed consent was obtained from all participants via survey checkbox.

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Availability of data and code statement

Data and code may be made available upon reasonable request.

CRediT authorship contribution statement

Kyle T. Ganson: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Alexander Testa:** Writing – review & editing, Methodology, Conceptualization. **Jason M. Nagata:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

All authors declare no conflicts of interest.

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